CAEM-DETR: Small Aerial Target Detection via Contrastive Attention-Enhanced Multidomain Fidelity Fusion

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***Abstract*—Small objects in uncrewed aerial vehicle (UAV) imagery, often vanishing amidst limited pixels, dynamic views, and complex backgrounds, present a critical detection challenge. State-of-the-art Transformer-based detectors, despite their strong general performance, frequently struggle to detect these small aerial targets, with their effectiveness noticeably decreasing. This article introduces contrastive attention-enhanced multidomain detection Transformer (CAEM-DETR), a novel framework derived from RT-DETR, specifically designed to overcome these intricacies and deliver significantly improved small object detection accuracy from aerial perspectives. CAEM-DETR incorporates several key innovations: 1) dynamic linear contrastive attention (DLCA), a mechanism that distinctively leverages positive and negative feature polarities, modeling their diverse interactions with linear complexity to enhance feature contrast and refine semantic representations for subsequent fusion; 2) multidomain fidelity fusion (MDFF), a module that synergizes high-fidelity downsampling with dual-domain (spatial and frequency) analysis, integrating DLCA-enhanced polarity information to recover critical high-frequency details from a comprehensive 4-D perspective; 3) GRACE-Net, a backbone employing synergistic multiattention (SMA) and selective channel modulation to capture fine-grained multiscale details; and 4) wise-FocalerIoU (WF), a novel composite loss function that provides robust, difficulty-aware gradient guidance for precise small object localization. Experimental results on four diverse datasets (VisDrone, AI-TOD, SIMD, and DOTA) demonstrate CAEM-DETR's effectiveness and generalization. On the VisDrone dataset, CAEM-DETR achieves a relative improvement of 13.7% in AP and 11.6% in AP_{50} over the baseline.**

***Index Terms*—Aerial imagery, AI-TOD, computer vision, deep learning, multidomain feature fusion, remote sensing, small object detection, Transformer, uncrewed aerial vehicles (UAVs), VisDrone.**

I. Introduction

THE widespread use and diverse applications of uncrewed aerial vehicles (UAVs), spanning military reconnaissance, disaster management, smart city infrastructure, and traffic monitoring [1], [2], [3], highlight the need for robust analytical capabilities, particularly in UAV-based object detection (UAV-OD). Consequently, UAV-OD has emerged as a significant research direction [4]. However, analyzing aerial imagery presents inherent challenges; targets are often small and densely distributed, viewpoints exhibit considerable variability, and backgrounds are frequently complex, collectively increasing the difficulty of detection tasks.

The advent of deep learning has substantially advanced object detection methodologies. Key paradigms include two-stage detectors like Faster R-CNN [5], which prioritize accuracy but with a higher computational cost; one-stage detectors such as YOLO [6] and RetinaNet [7], optimized for real-time processing but often struggling with small object precision [8]; and Transformer-based architectures. The detection Transformer (DETR) [9] marked a pivotal development by integrating Transformers with convolutional neural networks (CNNs), creating an end-to-end framework that eliminates the need for handcrafted components like nonmaximum suppression (NMS) and anchor boxes. Subsequent refinements, exemplified by RT-DETR [10], have further improved detection accuracy and convergence speed relative to traditional CNN-based methods.

However, despite notable progress in Transformer-based detectors like RT-DETR, their effectiveness in detecting small objects within aerial imagery is limited. For instance, while RT-DETR reports an AP_{50} of 65.1% on the MS COCO validation set [11], its performance markedly declines to 45.0% on the VisDrone2019-DET-val dataset [12], which is characterized by a large number of small targets. This performance disparity arises from several factors intrinsic to aerial perspectives; the limited pixel footprint of small objects restricts effective feature extraction and increases their susceptibility to interference from complex backgrounds and atmospheric conditions [13]. Existing frameworks often lack specialized mechanisms to adequately capture, distinguish, and

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utilize this fine-grained information. Furthermore, standard attention mechanisms can struggle to distinguish small targets from visually similar background clutter, and conventional loss functions, like GIoU [14], may lack the sensitivity needed for precise localization of such small objects. These combined factors severely hinder the reliable detection and representation of small, densely packed objects from aerial viewpoints.

To address the specific challenges of detecting small objects in aerial imagery, we introduce contrastive attention-enhanced multidomain DETR (CAEM-DETR), a novel framework with targeted improvements designed for small object detection. CAEM-DETR integrates strategic improvements across its architecture, specifically targeting the backbone network, feature fusion mechanisms, attention modules, and loss function design. Our key contributions are as follows.

1) We propose dynamic linear contrastive attention (DLCA), a mechanism that incorporates often overlooked information derived from negative feature polarities. It explicitly models interactions between features of both same and opposite polarities with linear computational complexity, dynamically enhancing feature contrast. This significantly refines high-level semantic representations to enhance subsequent multiscale fusion.
2) We design multidomain fidelity fusion (MDFF), a multiscale feature fusion module. Through high-fidelity downsampling and dual-domain processing that integrates large-kernel convolutions, it effectively preserves and fuses spatial and frequency information. Combined with the polarity information, MDFF recovers critical high-frequency details drawn from a 4-D perspective, essential for better distinguishing small targets within complex backgrounds.
3) We develop a gated representation with attention and cross-stage enhancement network (GRACE-Net), a backbone that captures fine-grained multiscale information through synergistic multiattention (SMA) mechanisms and selective channel modulation.
4) We formulate wise-FocalerIoU (WF), a novel composite loss function designed to provide robust gradient guidance for precise small object localization through difficulty-aware weight allocation and adaptive interval mapping.

II. Related Work

A. Object Detection in UAV Imagery

Object detection in UAV imagery presents unique challenges, including small object dimensions, varied viewpoints, complex backgrounds, and dense target distributions. These challenges require specialized approaches that balance detection accuracy with computational efficiency—a critical factor for deployment on resource-constrained UAV platforms.

Traditional UAV-OD approaches employed two-stage detection frameworks such as Faster R-CNN [5], which achieved high accuracy at the expense of computational efficiency. These frameworks often incorporated feature enhancement techniques like feature pyramid networks (FPNs) [15] to improve multiscale feature representation. However, their significant computational overhead made them unsuitable for UAV applications requiring real-time processing. Consequently, research shifted toward single-stage detectors to achieve better efficiency–accuracy tradeoffs. The YOLO [6] architecture family became popular in UAV-OD due to its balanced performance, with optimized variants such as SlimYOLOv3 [16] specifically addressing the computational constraints of UAV platforms.

Recent studies have addressed small object detection challenges through various strategies: implementing hierarchical refinement pipelines [17], [18], extracting more discriminative features from higher resolution feature maps [4], [19], and using contextual information to enhance detection capabilities [20], [21]. Despite these advances, most existing UAV-OD methods focus primarily on spatial domain feature extraction, leaving frequency-domain information largely unexplored. Furthermore, research on optimizing postprocessing techniques specifically for UAV-OD applications remains limited, creating opportunities for further architectural innovation.

B. Transformer-Based Object Detection

DETR [9] marked a significant transformation in the field by its synergistic integration of Transformers and CNNs, which formed a complete, end-to-end pipeline, obviating manually engineered stages like NMS and anchor box. However, despite demonstrating competitive results on standard benchmarks, the slow training process of DETR and its difficulties in effectively identifying small objects limited its practical deployment, particularly for UAV imagery containing numerous small targets.

Subsequent research addressed these limitations through architectural refinements. Deformable convolution operations [22] were a major advancement by enabling dynamically adjustable receptive fields that adapt to object geometry and orientation variations. Building upon this concept, Deformable DETR [23] introduced deformable attention to focus on sparse spatial locations, significantly improving convergence and performance on small objects. This approach leverages the flexibility of deformable sampling to enhance the Transformer's capability to handle objects with varying scales and shapes.

More recently, RT-DETR [10] emerged as the first real-time end-to-end detector, outperforming conventional YOLO models in both speed and accuracy through its use of attention-based intrascale feature interaction (AIFI), CNN-based cross-scale feature fusion (CCFF), and uncertainty-minimal query selection. RT-DETR's global attention mechanisms effectively capture long-range dependencies and contextual information, making it inherently more adaptable than traditional CNN-based detectors.

Despite these advancements, Transformer-based detectors still face significant challenges when applied to UAV imagery. As noted in the introduction, RT-DETR's performance differential—AP_{50} of 65.1% on MSCOCO-val [11] versus AP_{50} of 45.0% on VisDrone2019-DET-val [12]—highlights a substantial gap when detecting small objects in aerial imagery. This disparity underscores the need for specialized architectural components precisely tailored to address the unique characteristics of UAV-OD applications.

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C. Multiscale Feature Fusion Techniques

Object detection critically relies on feature fusion techniques for integrating information across multiple scales to improve the detection accuracy of targets of varying sizes. However, a significant challenge is the the semantic gap between features from disparate levels, particularly when identifying small and closely packed targets in UAV imagery.

Conventional approaches use direct methods such as element-wise summation or channel-wise concatenation of feature maps, though these often lead to spatial feature misalignment [24]. More sophisticated techniques include FPNs [15], path aggregation networks (PANet) [25], and bidirectional FPN (BiFPN) [26], which create pathways for bottom-up and top-down feature refinement. In the domain of remote sensing, where multisource and multiscale data are common, specialized fusion strategies have also been developed. For instance, to address the challenge of classifying imbalanced data with limited labels, Miao et al. [27] proposed a multigranularity decoupling network capable of achieving superior semi-supervised classification performance.

Attention-based fusion mechanisms [28] further address alignment issues but focus primarily on spatial feature fusion without fully leveraging frequency-domain information. While recent studies have begun to explore frequency-domain fusion techniques [29], [30], recognizing the complementary information available in this domain, many of these methods do not fully exploit its potential. Furthermore, they typically overlook the rich contrastive information embedded in feature polarities, particularly negative feature polarities. The integration of such multiscale information across these diverse dimensions is crucial for precisely distinguishing small targets within complex aerial imagery.

D. Specialized Approaches for Small Object Detection

In addition to multiscale feature fusion, various specialized techniques for small object detection have also emerged. Among these are cascaded refinement architectures, including Cascade R-CNN [31], [32] and QueryDet [18], designed to iteratively enhance localization accuracy. Other approaches prioritize feature enhancement specifically for small targets, exemplified by frameworks like UAV-YOLO [33].

Attention mechanisms represent another key strategy, enhancing feature extraction for small objects. The convolutional block attention module (CBAM) [28], for instance, sequentially applies channel and spatial attention to refine feature maps. Similarly, HRFormer [34] integrates local-window self-attention within a high-resolution framework to capture fine-grained details. Furthermore, leveraging high-resolution feature representations offers significant advantages. HRNet [35] maintains high-resolution pathways throughout the network via parallel multiresolution streams and frequent information exchange, yielding features that are both semantically rich and spatially precise. Building upon this concept, HRDNet [36] employs multiresolution inputs and multidepth backbones to preserve positional information while reducing computational overhead. Additionally, techniques using dilated or deformable convolutions provide adaptive receptive fields, enabling networks to better adjust to varying object scales and detect small targets [22], [37].

Despite these advancements, significant challenges remain, as many existing methods do not fully resolve the fundamental difficulties of small object detection. Key limitations include: the low feature discriminability for objects with a small pixel footprint; increased vulnerability to interference from complex backgrounds and noise; the limited sensitivity of standard attention mechanisms in differentiating small targets from clutter; and the suboptimal guidance from conventional loss functions for the precise localization of small objects. Moreover, standard evaluation metrics often fail to provide sufficient optimization gradients when minor localization errors drastically impact performance outcomes for small targets.

Our proposed CAEM-DETR framework is designed to directly address these ongoing challenges through an integrated approach. It incorporates targeted innovations across the backbone network architecture, attention mechanisms, feature fusion strategies, and loss function formulation. Each component has been specifically engineered to enhance small object detection performance, particularly within the context of UAV imagery, while concurrently striving for computational efficiency suitable for real-time deployment.

III. Methodology

A. Architecture Overview

Fig. 1 illustrates CAEM-DETR, our proposed framework built upon the RT-DETR [10] architecture—a leading end-to-end object detector known for its excellent balance of speed and accuracy (RT-DETR-r18 serves as our baseline). CAEM-DETR comprises four main modules: the foundational backbone network, an optimized integrated encoder, a decoder, and heads for prediction. The workflow begins with the backbone producing multiscale feature representations ($\{P_2, P_3, P_4,$ and $P_5\}$). These features are then fed into the fine-grained cross-scale feature fusion (FCFF) module, which integrates both intrascale and interactions between features. Next, leveraging an IoU-aware query selection strategy, salient image features are chosen as the initial inputs for the decoder. The decoder then works to iteratively enhance these queries, ultimately producing precise bounding boxes and associated confidence scores.

B. Backbone

Standard backbones often struggle to distinguish small, morphologically complex targets in UAV imagery, primarily due to the progressive loss of fine-grained details during downsampling and the limited receptive fields of conventional convolutions. To overcome these limitations, we introduce GRACE-Net. Its design philosophy is rooted in maximizing feature discriminability for small objects while carefully managing the computational budget. This is achieved through a strategic integration of two specialized blocks: the GRACE convolutional block (GCB) for efficient feature preservation

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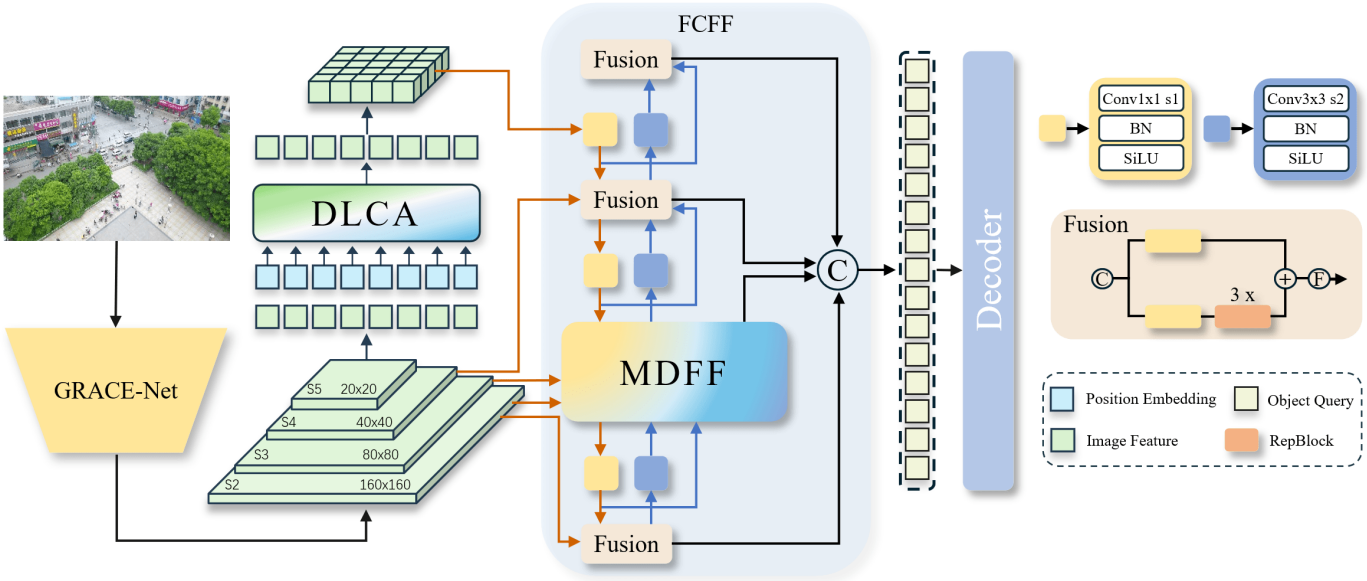


Fig. 1. Overview of the CAEM-DETR architecture. Our framework integrates GRACE-Net, DLCA, FCFF, and MDFF modules to enhance small target detection in aerial imagery.

and the GRACE attention-convolutional block (GACB) for advanced feature refinement.

The foundational GCB, illustrated in Fig. 2(a), introduces an efficient dual-stream information flow. Its operation begins with a 1×1 convolution to adjust channel dimensions, immediately followed by a split operation that partitions the feature map into two parallel streams. This "split-transform-merge" paradigm is computationally efficient. One stream is maintained as a short-cut connection, which is crucial for preserving high-fidelity spatial details and ensuring unimpeded gradient flowing through the network. The parallel stream employs a lightweight bottleneck structure with 3×3 convolutions to extract richer local patterns without incurring significant computational cost. By concatenating these two streams, GCB effectively fuses original, unmodified features with newly learned abstract representations, creating a robust feature map that retains critical detail.

Building upon this efficient foundation, the GACB modules [Fig. 2(b)] enhance feature representation by replacing the simple convolutional stream with a more powerful GRACE attention unit (GAU). This unit, depicted in Fig. 2(c) and inspired by SMAFormer [38], is purpose-built to capture the complex, long-range dependencies essential for contextualizing small objects. The GAU comprises two key innovations. The first is synergistic multiattention (SMA), detailed in Fig. 2(d). SMA moves beyond singular attention mechanisms by synergistically integrating channel, spatial, and multihead self-attention. This allows the model to capture true global context and understand the interdependencies between disparate regions in the feature map. The second innovation is the convolutional gated linear unit (CGLU) [39], as shown in Fig. 2(e). It serves as an adaptive, spatially aware gating mechanism. By leveraging efficient depth-wise convolutions (DWConvs) within its gating structure, CGLU can selectively modulate channel-wise feature responses based on their local

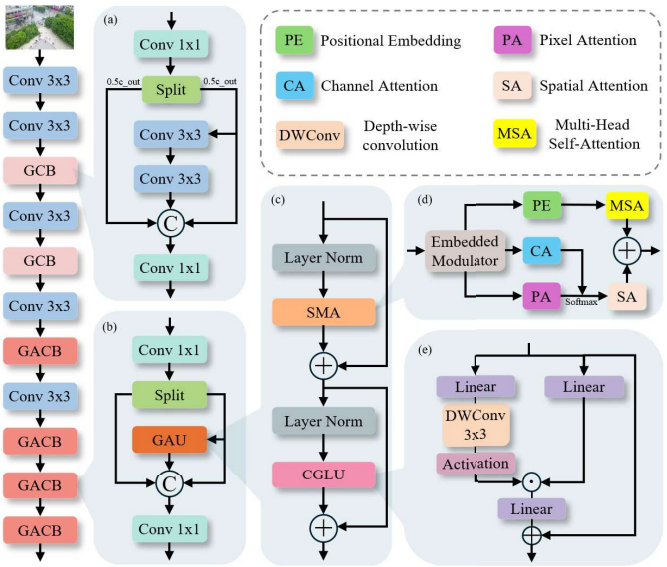


Fig. 2. GRACE-Net architecture showing (a) GCB with dual-path structure for local and global feature processing, (b) GACB incorporating attention mechanisms, (c) GAU integrating sophisticated attention, (d) SMA module fusing multiple attention types, and (e) CGLU implementing adaptive gating.

spatial context, effectively amplifying salient target features while suppressing background noise with minimal parameter overhead.

In essence, the hierarchical architecture of GRACE-Net—progressing from the detail-preserving efficiency of GCBs to the context-aware power of GACBs—creates a feature extraction pipeline that is both powerful and economical. It is precisely this combination of fine-grained detail preservation and global contextual reasoning, all achieved within a constrained computational framework, which provides a superior feature foundation for detecting small targets in challenging aerial scenes.

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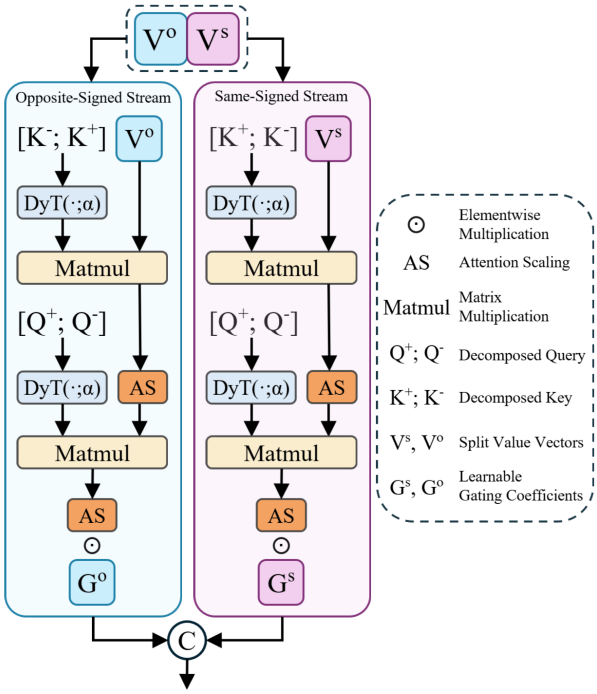


Fig. 3. Structure of the DLCA module. It illustrates the dual-stream architecture, with its same-signed stream and opposite-signed stream, that processes interactions using polarity-decomposed inputs. Key operations include the DyT transformation for adaptive feature scaling, AS blocks, and modulation by learnable, query-specific coefficients (G^s and G^o) prior to output concatenation.

C. Dynamic Linear Contrastive Attention

For small objects in aerial imagery, distinguishing features often become more apparent through contrast or dissimilarity with the background (i.e., opposite-polarity interactions), rather than solely through faint similarities. Standard attention mechanisms, such as the AIFI module in the RT-DETR baseline, tend to overlook the rich information embedded in negative query–key interactions. Furthermore, they often produce diffuse attention maps, struggle with the inherent variations of aerial environments, and incur high computational costs ($\mathcal{O}(N^2)$), making them inefficient for high-resolution inputs. Drawing inspiration from the principles of PolaFormer [40], which highlights the importance of feature polarity, and recent findings on normalization-free Transformers [41], we introduce the DLCA module to overcome these limitations.

As illustrated in Fig. 3, DLCA is engineered to consider a broader spectrum of feature interactions by explicitly modeling both positive and negative polarities. A key innovation within DLCA is the integration of the dynamic tanh (DyT) function. The motivation for DyT stems from the observation in [41] that standard layer normalization often imposes a tanh-like, S-shaped transformation on activations. DyT, defined as

$$\mathrm{DyT}(x) = \tanh(\alpha x) \tag{1}$$

serves as a lightweight, learnable alternative that mimics this behavior. It adaptively scales and squashes activations, which is crucial for enhancing feature contrast.

The core operation of DLCA involves decomposing query (Q) and key (K) vectors into their positive (Q^+ and K^+) and negative (Q^- and K^-) components. The value (V) vector is split into two halves, V^s and V^o. The module then uses a dual-stream architecture to separately process same-signed and opposite-signed interactions, ensuring a comprehensive modeling of feature relationships. The approximate attention output for a given query Q_t is formulated as

$$O_t^s = \frac{\sum_{i=1}^{N} \mathrm{Sim}^s(Q_t, K_i) \cdot \mathcal{A}_1(V_i^s)}{\sum_{j=1}^{N} \mathrm{Sim}^s(Q_t, K_j)} \odot G_t^s \tag{2}$$

$$O_t^o = \frac{\sum_{i=1}^{N} \mathrm{Sim}^o(Q_t, K_i) \cdot \mathcal{A}_1(V_i^o)}{\sum_{j=1}^{N} \mathrm{Sim}^o(Q_t, K_j)} \odot G_t^o \tag{3}$$

$$\mathrm{DLCA}(Q_t, K, V) \approx \mathrm{Concat}(O_t^s, O_t^o) \tag{4}$$

where V^s and V^o represent the partitioned value vectors for the same-signed and opposite-signed streams, respectively, and G_t^s and G_t^o are the corresponding learnable, query-specific modulation matrices. The similarity functions Sim^s and Sim^o incorporate the DyT transformation

$$\mathrm{Sim}^s(Q_t, K_i) = \mathrm{DyT}\left(\left[Q_t^+; Q_t^-\right]; \alpha\right) \cdot \mathrm{DyT}\left(\left[K_i^+; K_i^-\right]; \alpha\right)^\top \tag{5}$$

$$\mathrm{Sim}^o(Q_t, K_i) = \mathrm{DyT}\left(\left[Q_t^+; Q_t^-\right]; \alpha\right) \cdot \mathrm{DyT}\left(\left[K_i^-; K_i^+\right]; \alpha\right)^\top. \tag{6}$$

In these formulations, $[;]$ denotes the concatenation. The term $\mathcal{A}_1(\cdot)$ conceptually represents the initial attention scaling (AS) block depicted in Fig. 3, which operates on the value vectors within each stream. The complete computation leverages the associative property of matrix multiplication to achieve linear complexity ($\mathcal{O}(N)$) and includes additional AS blocks as shown in the figure.

By explicitly modeling the full spectrum of query–key interactions through both positive and negative feature polarities and by incorporating the adaptive DyT transformation, DLCA generates highly discriminative and focused attention maps. This approach significantly improves the model's ability to distinguish small objects from complex backgrounds in aerial imagery, while preserving computational efficiency.

D. Multidomain Fidelity Fusion

Effective detection of small targets in UAV-acquired scenes requires robust multiscale feature fusion approaches capable of bridging the semantic gap between high-resolution, detail-rich low-level features and low-resolution, semantically meaningful high-level features. Conventional downsampling operations, such as strided convolutions and pooling, while effective in expanding receptive fields and reducing computational load, often sacrifice critical fine-grained information, causing the features of small objects to be gradually lost in deeper network layers. Although certain methods, like SPDConv [42], aim to preserve image details during such downsampling, reliance solely on spatial domain information remains insufficient for robust small object detection in complex aerial scenes. Visually similar background elements can be easily confused with small targets in the spatial domain, creating a form of "natural camouflage." In such challenging scenarios,

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frequency-domain analysis offers a novel perspective: many background noise patterns or periodic textures exhibit distinct signatures in the frequency domain compared to the target signals. Effectively leveraging frequency information could enable the model to "look beyond" spatial domain ambiguities and capture critical differences between the target and background based on their respective frequency components. Inspired by advancements in multidomain feature representation, particularly concepts demonstrated in architectures like the Omni-Kernel Network [29], we propose the MDFF module to address these complex challenges. MDFF, a core component of our FCFF as depicted in the overall architecture (Fig. 1), initially employs space-to-depth convolution (SPDConv) to ensure the high-fidelity transmission of small object information, thereby providing an information-rich and clean feature foundation for subsequent processing. Building upon this, MDFF then synergistically integrates analyses from both spatial and frequency domains, guiding a deep and comprehensive fusion of features to enhance the discriminability of small targets.

As illustrated in Fig. 4, SPDConv rearranges spatial feature map blocks into the channel dimension, effectively downsampling the spatial resolution by a factor r while increasing the channel dimension by r^2, thereby preserving the complete information content present in the original feature map. This transformation converts the S2 feature map $F_{S2} \in \mathbb{R}^{H\times W\times C}$ into $F'_{S2} \in \mathbb{R}^{(H/r)\times(W/r)\times C\cdot r^2}$. The information preserving F'_{S2} is then concatenated with the features from the S3 stage, followed by channel reduction using 1×1 convolutions, preparing the features for multiscale integration.

Following the initial SPDConv-based enhancement and concatenation, MDFF employs a multibranch core structure inspired by the Omni-Kernel module (OKM) citecui2024omni to extract diverse and complementary features across local, large-scale, and global contexts (Fig. 5). The **Local Branch** uses a 1×1 DWConv which operates channel-wise to refine features locally without spatial mixing, preserving fine details crucial for distinguishing small objects. To capture contextual information across varying spatial extents and orientations, the **Large Branch** utilizes parallel DWConvs with exceptionally large and diverse kernel shapes: 63×63, 1×63, and 63×1; these large receptive fields, efficiently implemented using depth-wise operations, allow the network to model extensive spatial relationships relevant for contextualizing objects, even when placed in the bottleneck where feature maps are smaller. The **Global Branch** uses dual-domain processing to capture global dependencies, essential for understanding the broader scene context. It incorporates the dual-domain channel attention module (DCAM), which, as detailed in Fig. 5(a), combines frequency channel attention (FCA) via fast Fourier transform (FFT)/inverse FFT (IFFT) and spatial channel attention (SCA) using global average pooling (GAP) to adaptively modulate features based on global channel-wise statistics in both spatial and frequency domains, and the frequency-based spatial attention module (FSAM), shown in Fig. 5(b), which further refines features by applying spatial attention modulated in the frequency domain, allowing the model to focus on the most informative frequency components for reconstruction.

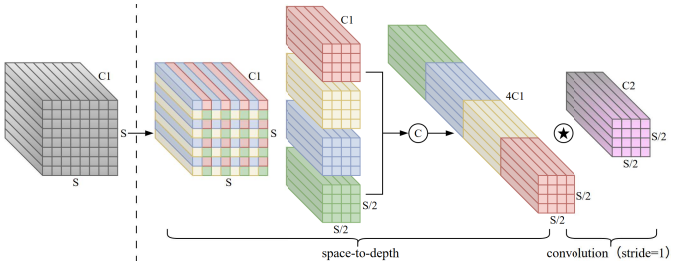


Fig. 4. Illustration of the SPDConv block, consisting of a SPD layer followed by a nonstrided convolution. This block performs information-preserving feature map downsampling.

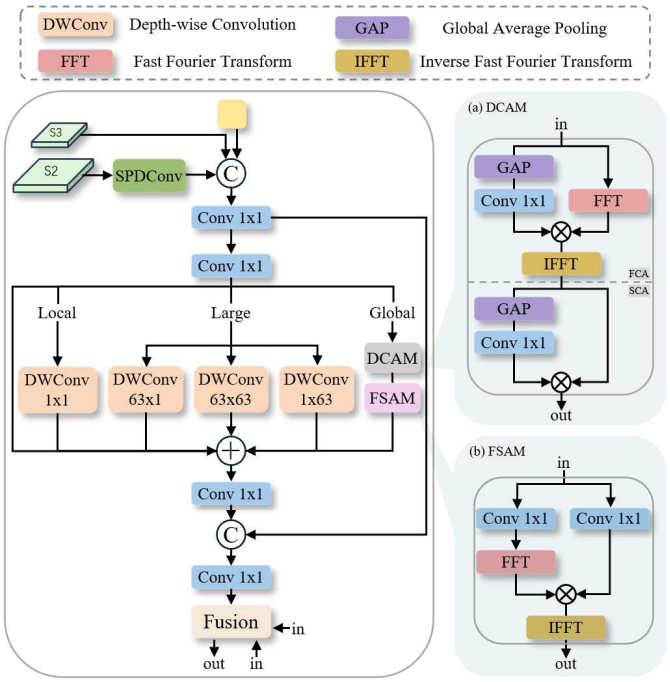


Fig. 5. Overview of the MDFF module. It uses initial SPDConv processing followed by a multi-domain architecture with Local, Large, and Global branches. The Global branch integrates two key components for feature enhancement. (a) Dual-domain channel attention module (DCAM). (b) Frequency-based spatial attention module (FSAM).

Features extracted by the local, large, and global branches are aggregated via element-wise addition. This fused representation is then processed by a 1×1 convolution. A skip connection concatenates this result with the output from the initial 1×1 convolution stage (following the SPDConv and S3 concatenation). Subsequent 1×1 convolutions and a final fusion block refine the output before it exits the MDFF module.

This hierarchical integration strategy, combining information-preserving downsampling via SPDConv with the multiscale, multidomain feature extraction capabilities of the Omni-Kernel core, enables MDFF to generate highly discriminative feature representations. By effectively preserving fine-grained details while capturing broad spatial and global context, MDFF significantly enhances the framework's ability to detect and precisely localize small objects within challenging UAV imagery. The output features from MDFF are then passed to subsequent fusion layers

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within the FCFF module for further integration across different network stages (Fig. 1).

E. WF Loss Function

To enhance bounding box regression (BBR) precision for small objects in UAV imagery, where standard approaches like GIoU loss [14] often struggle with providing precise gradient guidance and mitigating interference from low-quality samples, we propose WF. This composite loss function synergizes two advanced mechanisms: the dynamic nonmonotonic focusing of Wise-IoU [43], which excels at adapting to sample quality, and the targeted interval mapping of Focaler-IoU [44], which allows for focusing on specific sample difficulties. By integrating these, WF aims to leverage their complementary strengths, enabling them to synergistically enhance performance and address the limitations of using each mechanism alone.

Focaler-IoU first reconstructs the standard IoU using linear interval mapping to focus the loss calculation on a specific IoU range $[d, u]$, beneficial for targeting particular sample difficulties

$$\mathrm{IoU}_{\mathrm{Focaler}} = \mathrm{clamp}\left(\frac{\mathrm{IoU} - d}{u - d}, 0, 1\right). \tag{7}$$

The corresponding Focaler-IoU loss is given by

$$\mathcal{L}_{\text{Focaler-IoU}} = 1 - \mathrm{IoU}_{\mathrm{Focaler}}. \tag{8}$$

Wise-IoU's dynamic mechanism then adaptively reweights this Focaler-IoU loss based on its deviation from the running mean $\overline{\mathcal{L}_{\text{Focaler-IoU}}}$. This is achieved by calculating an outlier degree β and applying the Wise-IoU-v3 nonmonotonic focusing coefficient r

$$\beta = \frac{\mathcal{L}^{*}_{\text{Focaler-IoU}}}{\overline{\mathcal{L}_{\text{Focaler-IoU}}}} \tag{9}$$

$$r = \frac{\beta}{\delta \alpha^{(\beta - \delta)}} \tag{10}$$

where $\mathcal{L}^{*}_{\text{Focaler-IoU}}$ denotes a detached version of the loss and α and δ are hyperparameters.

The final WF loss integrates these components. It applies the **Wise-IoU dynamic scaling factor** r to a **base geometric loss term**. In our implementation, this base term is constructed using the Focaler-IoU loss $\mathcal{L}_{\text{Focaler-IoU}}$ combined with the distance attention component $\mathcal{R}_{\text{WIoU}}$

$$\mathcal{R}_{\text{WIoU}} = \exp\left(\frac{\rho^2\left(\mathbf{p}, \mathbf{p}_{\text{gt}}\right)}{\left(W_g^2 + H_g^2\right)^{*}}\right) \tag{11}$$

$$\mathcal{L}_{\text{Wise-FocalerIoU}} = r \cdot \left(\mathcal{R}_{\text{WIoU}} \cdot \mathcal{L}_{\text{Focaler-IoU}}\right) \tag{12}$$

where $\rho^2(\mathbf{p}, \mathbf{p}_{\text{gt}})$ is the squared center distance and $(W_g^2 + H_g^2)^{*}$ is the squared diagonal of the smallest enclosing box (detached).

This formulation provides more refined gradient guidance by focusing on a relevant difficulty range (via Focaler-IoU) while dynamically downweighting outliers within that range using the Wise-IoU mechanism, enhancing localization accuracy for small objects in UAV scenarios.

IV. Experiments

A. Experimental Setup

1) Datasets: We conducted detailed quantitative evaluations on four established benchmarks for aerial object detection. Our primary experiments were conducted on VisDrone2019-DET [12], which contains 6471 training images, 548 validation images, and 3190 test images captured by UAVs across various altitudes and settings. To rigorously assess the generalization ability of our framework, we extended our evaluation to three additional and diverse datasets. The first, AI-TOD [45], is a demanding benchmark specifically designed for tiny object detection, with an average object size of just 12.8 pixels. We further validated our model on SIMD [46], a satellite imagery dataset for multiclass and multisized vehicle detection, and DOTA [47], a large-scale and challenging benchmark for object detection in aerial images. For all datasets, we followed their respective official protocols for training and evaluation.

2) Implementation Specifics: An NVIDIA GeForce RTX 4090 GPU served as the hardware platform for all experiments. CAEM-DETR is built upon the RT-DETR [10]. A training duration of 300 epochs was set, with a batch size of 4, without early stopping. For optimization, AdamW [48] was chosen, with its learning rate set to 1×10^{-4} and momentum to 0.9. All input images were resized to 640×640 dimensions. The standard data augmentation methods from the baseline RT-DETR implementation were applied for consistency. Additionally, a few additional augmentations from the `albumentations` library were used with a probability ($p = 0.01$): `Blur` and `MedianBlur` [with `blur_limit` set to (3, 7)], conversion to grayscale via `ToGray`, and `CLAHE` [with parameters `clip_limit` = (1.0, 4.0) and `tile_grid_size` = (8, 8)]. Performance was evaluated using standard COCO metrics: mean average precision (AP) averaged over a range of IoU thresholds from 0.50 to 0.95 (in 0.05 increments), as well as AP_{50} (denoting AP at an IoU of 0.50), AP_{75} (for AP at an IoU of 0.75), and AP scores for small (AP_S), medium (AP_M), and large (AP_L) object instances.

B. Comparative Analysis

As presented in Table I, CAEM-DETR demonstrates a significant performance leap over its RT-DETR-R18 baseline on the VisDrone validation set. This substantial gain, reflected in a 3.7 absolute point increase in AP (a 13.7% relative improvement) and a 5.2 point increase in AP_{50} (an 11.6% relative improvement), is a direct result of the synergistic integration of our proposed components which fundamentally enhance the model's feature representation and localization capabilities for challenging aerial targets.

A deeper analysis reveals the specific strengths of our approach, particularly in the context of small object detection. The notable improvement in this area, evidenced by the high AP_S score of 23.0, can be directly attributed to our core innovations. The MDFF module excels at preserving and fusing the high-frequency details crucial for distinguishing small targets, which are often lost in standard downsampling. Concurrently, the DLCA module enhances the contrast of these targets

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TABLE I
COMPARISON OF DIFFERENT OBJECT DETECTION METHODS ON THE VISDRONE2019-DET-VAL DATASET

Method	AP	AP_{50}	AP_{75}	AP_S	AP_M	AP_L	Params (M)	GFLOPs	FPS
Faster R-CNN [5] (2015)	23.6%	39.1%	24.7%	14.8%	34.8%	40.8%	41.4	208	55
Cascade R-CNN [31] (2018)	23.6%	38.5%	25.3%	14.8%	35.3%	41.8%	69.3	236	45
RetinaNet [7] (2017)	18.0%	30.2%	18.9%	8.2%	30.4%	37.2%	36.5	210	63
GFL [49] (2020)	22.6%	37.6%	23.0%	13.5%	34.2%	39.4%	32.3	206	61
TOOD [50] (2021)	23.7%	39.2%	24.4%	14.7%	35.0%	42.6%	32.0	199	44
YOLOv8-L [51] (2023)	26.6%	43.5%	27.4%	16.6%	39.8%	47.3%	43.6	165	**173**
YOLOv10-L [52] (2024)	26.5%	43.5%	27.0%	16.5%	39.3%	47.4%	25.7	126	150
YOLOv11-L [53] (2024)	26.2%	42.9%	26.9%	16.4%	38.8%	48.5%	25.3	87	45
YOLOv12-L [54] (2025)	26.4%	43.3%	27.0%	16.4%	39.4%	47.3%	26.3	89	42
RT-DETR [10] (2024)	27.1%	45.0%	27.5%	19.1%	37.1%	42.5%	**19.9**	**57**	90
Deformable DETR [23] (2020)	27.1%	42.2%	26.1%	19.6%	34.4%	40.2%	40.0	173	29
Sparse DETR [55] (2021)	27.3%	42.5%	26.4%	20.2%	35.0%	40.2%	41.0	121	35
Van-DETR [56] (2025)	28.1%	47.6%	–	–	–	–	38.0	127	96
UAV-DETR [57] (2025)	29.8%	48.8%	–	–	–	–	20.0	77	54
DINO [58] (2022)	29.4%	46.2%	29.1%	22.9%	38.4%	**44.5%**	47.6	279	21
CAEM-DETR (ours, 2025)	**30.8%**	**50.2%**	**31.8%**	**23.0%**	**40.7%**	42.5%	**19.9**	93	65

TABLE II
PERFORMANCE COMPARISON ON THE AI-TOD DATASET

Method	AP	AP_{50}	AP_S	AP_M
Faster R-CNN [5] (2015)	11.3%	25.4%	9.5%	36.4%
Cascade R-CNN [31] (2018)	13.5%	29.1%	11.7%	37.9%
YOLOv12-L [54] (2025)	23.1%	51.1%	21.9%	**42.9%**
HANet [59] (2023)	22.1%	53.7%	**27.3%**	36.8%
Focus-DETR [60] (2023)	16.7%	41.6%	–	–
Crater-DETR [61] (2024)	18.1%	42.1%	19.7%	–
RT-DETR [10] (2024)	20.9%	49.5%	20.0%	35.1%
CAEM-DETR (Ours, 2025)	**25.6%**	**56.7%**	24.9%	37.8%

TABLE III
GENERALIZATION PERFORMANCE COMPARISON ACROSS MULTIPLE DATASETS

Dataset	Method	AP	AP_{50}	AP_S
VisDrone [12]	RT-DETR [10] (Baseline, 2024)	27.1%	45.0%	19.1%
	CAEM-DETR (Ours, 2025)	**30.8%**	**50.2%**	**23.0%**
AI-TOD [45]	RT-DETR [10] (Baseline, 2024)	20.9%	49.5%	20.0%
	CAEM-DETR (Ours, 2025)	**25.6%**	**56.7%**	**24.9%**
DOTA [47]	RT-DETR [10] (Baseline, 2024)	37.5%	63.5%	23.8%
	CAEM-DETR (Ours, 2025)	**38.6%**	**65.3%**	**25.7%**
SIMD [46]	RT-DETR [10] (Baseline, 2024)	60.7%	76.3%	22.1%
	CAEM-DETR (Ours, 2025)	**62.8%**	**78.6%**	**28.6%**

against complex backgrounds by explicitly modeling a richer spectrum of feature interactions. This architectural superiority allows CAEM-DETR to outperform various YOLO-series detectors in overall accuracy under a comparable computational budget (<100 GFLOPS). While models like YOLOv8-L are highly optimized for general-purpose speed, CAEM-DETR offers a more compelling accuracy-efficiency tradeoff for the specific, high-precision demands of aerial imagery analysis.

The generalizability and robustness of our design are further validated on the demanding AI-TOD dataset, as shown in Table II. AI-TOD is characterized by an abundance of extremely small objects, with an average size of just 12.8 pixels, making it a powerful test of a model's fundamental capabilities. The competitive performance of CAEM-DETR in this scenario underscores that our architectural principles are not merely tuned for one dataset but are fundamentally effective for the broader problem of tiny object detection.

While CAEM-DETR excels in its primary goal of small object detection, a complete performance profile also involves considering the design tradeoffs inherent in such a specialized architecture. For instance, regarding inference speed, models like YOLOv8-L achieve a higher FPS. Our design deliberately prioritizes detection accuracy and robustness in complex scenes, representing a tradeoff between precision and raw throughput. Similarly, an analysis of performance across object scales shows that while our method yields substantial gains for small (AP_S) and medium (AP_M) objects, its performance on large objects (AP_L) is comparable to the baseline but does not surpass some other methods. This indicates that our optimizations are highly targeted and effective for our stated problem. These considerations are important for practical deployment, highlighting both the specific scenarios where CAEM-DETR provides the most significant advantages and potential avenues for future research.

C. Generalization Ability Evaluation

To rigorously evaluate the generalization ability and robustness of our proposed framework, we extended our experiments to two additional public benchmarks for aerial imagery: SIMD and DOTA. These datasets were chosen to verify that the performance advantages of CAEM-DETR are broadly applicable and not confined to our initial testbeds.

The results, summarized in Table III for a holistic comparison across all four datasets, show a clear and consistent trend; CAEM-DETR outperforms the RT-DETR-R18 baseline in every case. Crucially, the most significant performance gains are observed in the small object detection metric (AP_S), with increases of +3.9, +4.9, +6.5, and +1.9 points on VisDrone, AI-TOD, SIMD, and DOTA, respectively. This robust and

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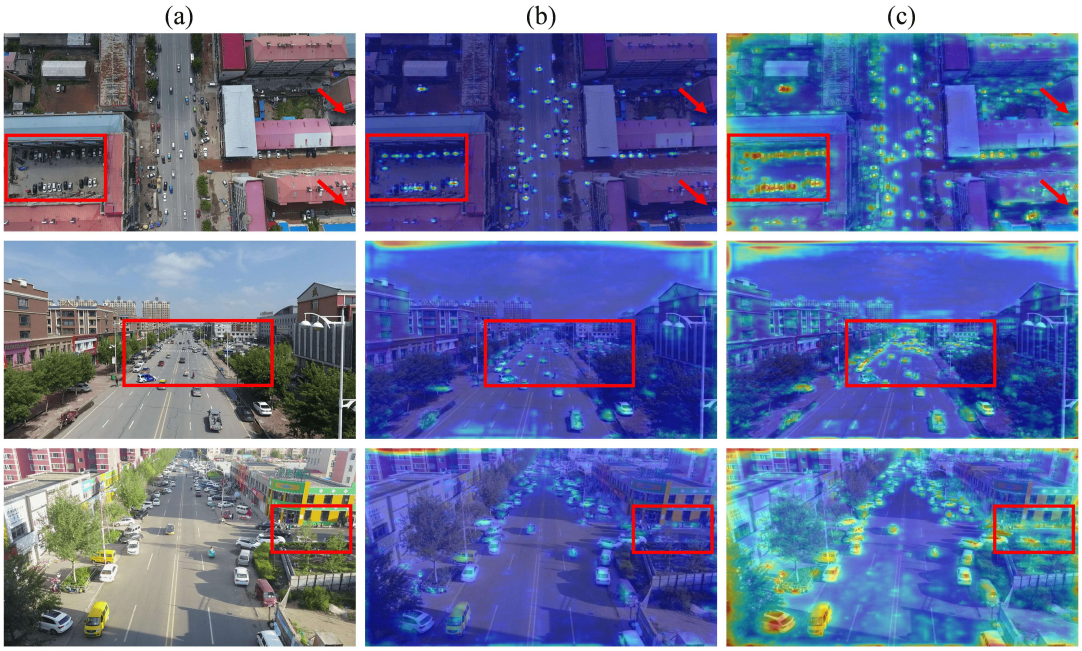


Fig. 6. Heatmap visualizations comparing the attention mechanisms. (Row) For each scene, the columns display (a) original input image, (b) activation map from the baseline RT-DETR, and (c) activation map from our CAEM-DETR. Brighter regions indicate stronger model attention. The comparison highlights that our model achieves a more intense and precise focus on targets while also leveraging broader contextual information, particularly in complex scenarios with small, clustered, and occluded objects (e.g., red box example).

TABLE IV
ABLATION STUDY ON THE VISDRONE2019-DET-VAL DATASET

GRACE-Net	MDFF	DLCA	WF	AP	AP_{50}	Params (M)
				27.1%	45.0%	19.9
✓				27.8%	45.6%	20.9
✓	✓			30.0%	48.7%	**19.7**
✓	✓	✓		30.5%	49.7%	19.9
✓	✓	✓	✓	**30.8%**	**50.2%**	19.9

transferable advantage provides strong evidence that our core innovations—such as the contrast enhancement from DLCA and the high-frequency detail preservation from MDFF—are fundamentally effective and generalize well to diverse aerial scenes.

D. Ablation Studies

We conducted a rigorous examination of the specific contributions from our proposed modules through extensive ablation studies using the VisDrone2019-det-val dataset. Table IV summarizes the results of this investigation.

Starting from the RT-DETR-R18 baseline performance of 27.1% AP and 45.0%AP_{50}, we incrementally integrated our proposed components. Replacing the original backbone with GRACE-Net first improved performance, yielding an AP of 27.8% (+0.7 AP). While this standalone replacement results in a marginal parameter increase of 1.0 M, the significant accuracy gain demonstrates a compelling performance-to-cost ratio. Furthermore, the true efficiency of our design is holistic; the synergy with our subsequent, highly parameter-efficient modules allows the final CAEM-DETR model to achieve its substantial performance leap while returning to the baseline's exact parameter count of 19.9 M. This underscores that the efficiency of our framework lies in maximizing accuracy without inflating the final model size.

Subsequently, incorporating the MDFF module for multiscale feature fusion significantly boosted the AP further to 30.0% (+2.2 AP over the previous step), highlighting the substantial impact of preserving fine-grained details and integrating dual-domain information; this component provided the largest single performance gain in our study. Adding the DLCA attention mechanism on top of this resulted in an AP of 30.5% (+0.5 AP) and notably improved the AP_{50} to 49.7% (+1.0 AP_{50}), suggesting enhanced localization precision through better semantic feature refinement. Finally, employing our proposed WF loss function achieved the best overall results, leading to the final model performance of 30.8% AP (+0.3 AP) and 50.2%AP_{50} (+0.5 AP_{50}).

The results validate the functionality of each individual module and underscore their synergistic contributions to the final performance of CAEM-DETR. Additional parametric studies on the WF loss, presented in Tables V and VI, confirmed that the combination of WiseIoU and FocalerIoU with parameters $d = 0.0$ and $u = 0.95$ yields optimal results for this task compared to other IoU loss variants and parameter settings.

E. Visualization Analysis

Fig. 6 presents the qualitative results via heatmap visualizations, focusing on activation patterns related to the predicted bounding boxes for small targets within the VisDrone dataset. A comparison of the heatmaps generated by the foundational

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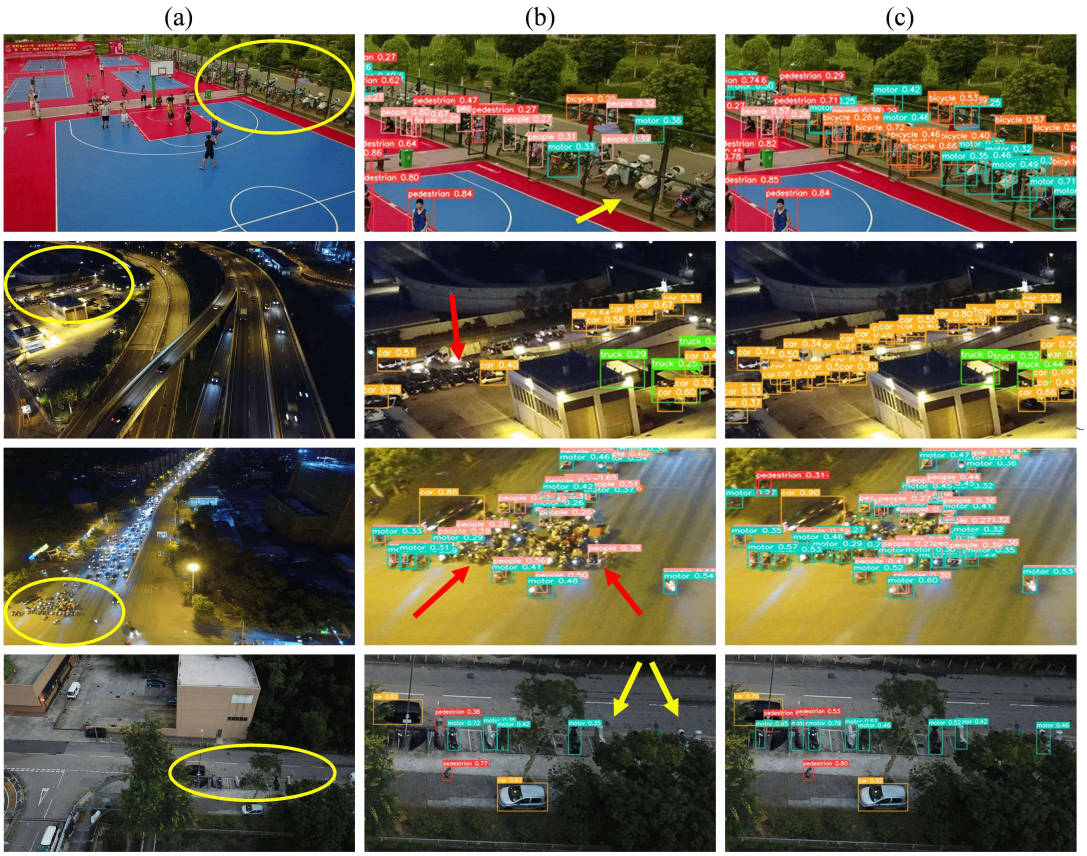


Fig. 7. Qualitative comparison on challenging VisDrone scenes. Each row displays (a) full scene with a highlighted region of interest, (b) baseline's (RT-DETR) result in the magnified region, and (c) our model's (CAEM-DETR) result. Arrows in (b) pinpoint missed detections in dense (red) or occluded (yellow) areas, which are successfully identified by our model in (c).

TABLE V
COMPARISON OF DIFFERENT IOU LOSS COMBINATIONS ON DETECTION PERFORMANCE

GIoU	WiseIoU	FocalerIoU	ShapeIoU	MPDIoU	AP	AP_{50}
✓					30.5%	49.7%
	✓				30.5%	49.6%
		✓			30.6%	49.7%
	✓	✓			**30.8%**	**50.2%**
	✓	✓	✓		30.6%	49.8%
	✓	✓		✓	30.1%	49.4%

TABLE VI
PARAMETRIC STUDY OF WISE-FOCALERIOU ON THE VISDRONE2019-VAL SET

d	u	AP	AP_{50}	AP_S	AP_M
0.0	0.95	**30.8%**	**50.2%**	**23.0%**	40.7%
0.0	0.90	30.4%	49.9%	22.7%	40.4%
0.05	0.95	30.3%	49.8%	22.2%	40.6%
0.05	0.90	30.6%	49.8%	22.6%	**40.8%**
0.1	0.90	30.5%	50.0%	22.8%	40.6%
0.2	0.90	30.4%	49.7%	22.6%	40.7%

RT-DETR-R18 and our CAEM-DETR reveals two fundamentally distinct attention strategies.

The baseline model exhibits a sparse and highly localized activation pattern. Its attention is confined to small, isolated regions, resulting in a visually darker background. This approach, however, reflects a limited contextual understanding, making the model vulnerable to errors when a target's intrinsic features are ambiguous, such as in cases of occlusion or low resolution.

In stark contrast, CAEM-DETR demonstrates a more sophisticated, dual-component attention mechanism. First, it achieves a significantly more intense and precisely focused activation peak directly upon the small targets, indicating a superior capability for feature discrimination. Second, and importantly, this sharp focus is supported by a broader field of contextual awareness, where the model actively engages with salient features in the target's vicinity. It is this active analysis of the surrounding environment—a process crucial for robust detection—that manifests as brighter background regions compared to the baseline.

This behavior is an intended outcome of our architecture's design. The DLCA module enhances feature contrast by inherently analyzing a target in relation to its local surroundings. This process is then amplified by the MDFF module, which uses its large receptive fields to integrate these locally aware features into a broader understanding of the scene. The resulting heatmaps, therefore, visualize a powerful synergy: a sharp, discriminative focus on the target, which is validated and supported by intelligent reasoning about its context. The practical impact of this advanced reasoning is visually

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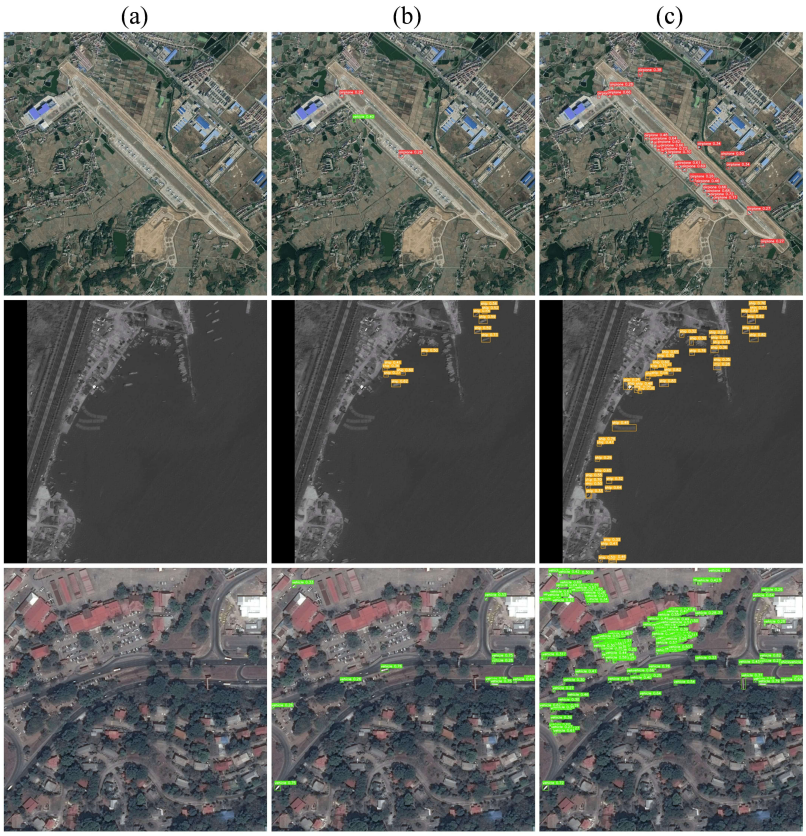


Fig. 8. Comparison of detection recall in large-scale, high-density AI-TOD scenes. (a) Original image, (b) sparse detections from the baseline RT-DETR, and (c) comprehensive detections from our CAEM-DETR. The visualization highlights our model's superior ability to identify a substantially larger number of densely packed objects missed by the baseline.

demonstrated in the detailed detection comparisons presented in Figs. 7 and 8.

Fig. 7 provides a detailed comparison in challenging scenes from the VisDrone dataset. Using a "magnifying glass" layout, it focuses on critical regions where the baseline model struggles. The visualizations, annotated with arrows, explicitly highlight instances of missed detections in dense crowds and under partial occlusion. In each case, CAEM-DETR consistently identifies these difficult targets, showcasing the tangible benefits of its design for robust and precise localization. Furthermore, Fig. 8 demonstrates the model's superior recall and generalization on large-scale, high-density images from the AI-TOD dataset. The side-by-side comparison starkly contrasts the sparse detections of the baseline with the comprehensive results from our model. This visual evidence confirms CAEM-DETR's effectiveness in identifying a substantially larger number of small, densely packed objects, underscoring its suitability for large-area remote sensing analysis.

F. Discussion

The experimental results demonstrate that CAEM-DETR offers compelling advantages for UAV-OD. First, inheriting the DETR paradigm, it eliminates the requirement for traditional hand-designed components, like NMS and anchor boxes, simplifying the detection pipeline. Second, its novel components strategically integrate and collaboratively process information from four dimensions—positive and negative feature polarities, as well as the spatial and frequency domains—which significantly enhances feature representation quality, particularly for small objects. Quantitative evaluations across multiple challenging aerial datasets (referenced in Tables I–III) confirm that CAEM-DETR delivers leading-edge results when compared with other detection models with comparable computational budgets (GFLOPs < 100) while maintaining inference speeds suitable for near real-time applications.

The observed performance gains stem from two primary synergistic innovations as follows.

1) *DLCA Attention:* It improves semantic feature alignment and discriminability by modeling both same-sign and opposite-sign feature interactions with linear complexity, allowing better differentiation of targets from cluttered backgrounds.
2) *MDFF Fusion Module:* It enhances multiscale feature integration by explicitly retaining high-frequency information and synthesizing spatial and frequency-domain representations through its multibranch, dual-domain architecture. Its approach of interweaving information from four dimensions allows it to overcome the limitations inherent in both purely spatial and traditional dual-domain fusion methods.

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V. Conclusion

This article introduced CAEM-DETR, an enhanced framework for small object detection guided by contrastive attention and multidomain fusion. By systematically addressing critical challenges—notably background interference, suboptimal feature utilization, and inadequate loss functions—our methodology significantly improves the ability to distinguish small objects against challenging aerial backgrounds. The synergistic integration of the DLCA and MDFF modules pioneers a novel 4-D feature processing strategy. This strategy leverages the contrast between positive and negative feature polarity dimensions to enhance subsequent dual-domain (spatial and frequency) fusion, enabling CAEM-DETR to effectively capture both local details and global context. Consequently, this approach overcomes the limitations of traditional methods, which often prioritize spatial domain analysis and rely mainly on positive feature characteristics. Within this framework, the MDFF module delivered the most substantial independent performance gain. This was complemented by the fine-grained multiscale detail capture from GRACE-Net and the precise gradient guidance from the WF loss, which further improved feature discrimination and refinement throughout the detection pipeline. Evaluations on four distinct datasets—VisDrone, AI-TOD, SIMD, and DOTA—confirm that CAEM-DETR achieves higher accuracy than existing methods under comparable computational constraints while also attaining near real-time inference speeds.

Future work will include refining feature extraction to reduce false positives from visually similar background elements and exploring domain adaptation techniques for better generalization. Furthermore, we plan to explore multimodal fusion, such as combining RGB with thermal or LiDAR data. This could provide complementary cues to improve detection robustness, especially for objects that are visually camouflaged or in adverse lighting conditions, directly addressing the core challenges of detection in complex aerial scenes. Integrating temporal information for video-based aerial detection is also planned.

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